

Kyung Ryun (Kenny) Kim

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EDUCATION

University of California, Irvine

Irvine, CA

Master of Science in Computer Science

Sep. 2026 – Jun 2028 (expected)

University of California, Irvine

Irvine, CA

Bachelor of Science in Software Engineering

Sep. 2023 – Jun 2026

EXPERIENCE

Undergraduate Research Assistant

April 2025 – Present

University of California, Irvine

Irvine, CA

- Developed a source-to-source C-to-Rust translation framework focusing on ownership and mutability inference
- Designed the high-level architecture for mutability analysis across MIR/HIR representations
- Implemented constraint-based mutability inference using Rust MIR/HIR to classify variable mutability

Undergraduate Research Assistant

June 2024 – Aug 2024

Univeristy of Osaka

Osaka, Japan

- Conducted research on testing and evaluating LLM-based software development workflows
- Analyzed LLM-assisted development under the waterfall model to identify reliability and testing challenges
- Presented “Application of MetaGPT to Software Development” at university’s academic seminar

PROJECTS

SSA IR for Tiny language | *Rust*

Jan, 2026 – March, 2026

- Designed and implemented SSA IR construction with automatic phi-function insertion at control flow join points
- Built recursive descent parser handling full expression grammar, function declarations, and nested control flow
- Implemented common subexpression elimination and copy propagation optimizations using per-block instruction storage tables

Sound Ethics Capstone Project | *Python, FastAPI, TypeScript, React, Google Cloud*

Sep, 2025 – March, 2026

- Implemented secure authentication and (single/multiple) audio file upload workflows using RESTful APIs
- Designed and deployed PostgreSQL schemas (users, audio files, folders) on Google Cloud SQL

Crux-To-x86 Compiler | *Java, GitLab*

Jan, 2025 – March, 2025

- Designed a typed intermediate representation (IR) bridging high-level Crux semantics and x86 assembly
- Lowered control flow, expressions, and function calls into IR for analysis and code generation
- Generated x86 assembly with stack frame management and register allocation

Operating System in Rust | *Rust*

June, 2025 – Aug, 2025

- Implemented core OS components including interrupt handling, virtual memory subsystem, and heap allocation
- Gained hands-on experience with low-level systems programming and hardware-software interfaces

Sport Scout App Generator | *Python, MCP, LangGraph, Gemini API*

Nov, 2025 – Dec, 2025

- Architected a multi-agent orchestration pipeline using MCP for automated code analysis, generation, and testing
- Implemented analyzer-developer-tester agent feedback loops to iteratively refine generated applications

TECHNICAL SKILLS

Programming Language: Java, Python, C/C++, Rust, JavaScript, TypeScript

Frameworks/Libraries: React, Angular, Django, JUnit, Selenium

Database: PostgreSQL, Google Cloud SQL, AWS DynamoDB

Tools: Git, Docker, Z3